

Program: BTech (CSE) Stream: CyberSecurityYear: 2022-23 Semester: VISubject: APPLICATION SECURITY TESTINGTime: 3 hrs (2 to 5pm)Date: 24 / April / 2023No. of Pages: 02Marks: 100**Final Examination**

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) **In all 5 questions to be attempted.**
- 4) All questions carry equal marks.
- 5) **Answer to each new question to be started on a fresh page.**
- 6) **Figures in brackets on the right hand side indicate full marks.**
- 7) **Assume Suitable data if necessary.**

Q1		Answer briefly:	
CO-1; SO-1; BL-2; U1	a.	Infer the need for securing information and comment on any TWO Information Security Attack Vectors.	[05]
CO-1; SO-1; BL-2; U2	b.	Describe the SAMM Framework for assessing the Secure Software Lifecycle.	[05]
CO-3; SO-4; BL-3; U3	c.	Comment on the reasons for Race Condition Vulnerability.	[05]
CO-2; SO-1; BL-2; U5	d.	Infer and describe the technique for stealing information from mobile device via Bluetooth.	[05]
Q2 CO-1; SO-1; BL-4; U1	a.	Working and Significance of Serverless Architecture compared with Single-page Architecture.	[10]
Q2 CO-2; SO-1; BL-1; U4	b.	Specify the three types of XSS attacks and specify the distinct steps of their working?	[10]

Q3 CO-2; SO-1; BL-3; U3	a.	Comment on Static Detection techniques and relate to how and which vulnerability it helps in detecting.	[10]
Q3 CO-3; SO-4; BL-4; U4	b.	Difference between CSRF and SSRF attacks wrt their working.	[10]
Q4 CO-1; SO-1; BL-2; U2	a.	Infer the need for adaptation of SecSDL to mobile environment.	[10]
Q4 CO-2; SO-1; BL-1; U5	b.	Explain Simjacker and Agent Smith Attack and its working.	[10]
Q5 CO-2; SO-1; BL-2; U3	a.	How would runtime detection help in mitigating structured output generation vulnerabilities.	[10]
Q5 CO-3; SO-1; BL-1; U5	b.	Describe general guidelines for Mobile Platform Security.	[10]
Q6 CO-1; SO-1; BL-4; U1	a.	Compare between SAST, DAST and IAST application security tools and solutions.	[10]
Q6 CO-2; SO-1; BL-1; U4	b.	Explain the FIVE defenses wrt preventing SQLi Vulnerabilities.	[10]
Q7 CO-2; SO-1; BL-1; U2	a.	Explain the SDL practices incorporated by Microsoft to improve the security of their products.	[10]
Q7 CO-3; SO-4; BL-4; U4	b.	Differentiate OAuth with SAML and describe the working of OAuth.	[10]